

SDR-Based Implementation and Metrological Microwave Characterization of Fractal NLFM Radar Waveforms up to 5.8 GHz

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Abstract—Non-Linear Frequency Modulated (NLFM) radar waveforms provide significant advantages over conventional Linear Frequency Modulation (LFM) by achieving low Peak Sidelobe Ratios (PSLR) without mismatched weighting filters. However, classical analytical NLFM designs (such as tangent, sinusoidal, or polynomial chirps) suffer from incomplete stationary phase synthesis and high near-in sidelobes (> -37 dB), while conventional numerical windowing (such as Kaiser-POSP) lacks native conjugate polarimetric architectures. In this paper, we present the mathematical derivation, analytical sidelobe cancellation theory, and traceable microwave metrological characterization of the Fractal Oscillation Source Model (FOSM) synthesized on the industrial PlutoSky SDR platform (AD9361 / Zynq-7020) operating in C-band at $f_c = 5.800$ GHz. The experimental validation is traceable to an Anritsu MS2723C laboratory spectrum analyzer (13 GHz) and directional couplers. We demonstrate analytically and experimentally that the proposed FOSM waveform maintains a strictly constant 0 dB Peak-to-Average Power Ratio (PAPR), achieves 0 dB SNR mismatch loss, delivers a theoretical PSLR of **-52.76 dB** (confirmed at **-47.53 dB** in traceable 24-bit audio metrology and **-45.05 dB** in 12-bit 5.8 GHz RF loopback, delivering a +27.71 dB gain over measured RF LFM), and enables time-reversed conjugate waveform pairs for simultaneous transmit and receive (STAR) polarimetric radar with cross-channel isolation exceeding > 30.5 dB (in strict agreement with the asymptotic $1/\sqrt{BT}$ bound).

Index Terms—Radar waveforms, Non-Linear Frequency Modulation (NLFM), Software-Defined Radio (SDR), Metrology, Pulse Compression, Waveform Diversity, Fractal Oscillation Source Model (FOSM), AD9361, Microwave Radar.

1 Introduction

Pulse compression radar systems rely on wideband frequency modulations to decouple range resolution from pulse duration [1], [2]. While Linear Frequency Modulation (LFM) is standard across military and civilian radar platforms, its intrinsic high auto-correlation sidelobes (-13.26 dB) necessitate amplitude windowing filters at the receiver (such as Hamming or Chebyshev windows). This conventional approach incurs an inevitable Signal-to-Noise Ratio (SNR) mismatch loss ($L_{\text{SNR}} \approx 1.3$ to 1.8 dB) and substantial mainlobe broadening.

Non-Linear Frequency Modulation (NLFM) overcomes this limitation by synthesizing a tapered energy spectral density directly through non-uniform time-frequency stationary phase dwell times [3], achieving low sidelobes with zero SNR mismatch loss ($L_{\text{SNR}} = 0$ dB). In modern Solid-State Transmitters (SST) based on Gallium Nitride (GaN) and Gallium Arsenide (GaAs) Power Amplifiers, operating with strictly constant envelope (0 dB PAPR) is mandatory to achieve maximum power-added efficiency (PAE) in saturated classes without non-linear AM-AM/AM-PM distortion. However, classic analytical closed-form NLFM functions (such as Tangent, S-curve, or Polynomial models) fail to conform the spectrum optimally, while numerical stationary phase windows (such as Taylor-POSP or Kaiser-POSP) suffer from edge derivative discontinuities under physical quantization. Modern Software-Defined Radio (SDR) platforms offer unprecedented agility to synthesize such arbitrary phase trajectories directly in baseband [4].

In this work, we present the comprehensive mathematical formulation, analytical destructive sidelobe cancellation derivation, and traceable microwave laboratory metrology of the Fractal Oscillation Source Model (FOSM) up to 5.800 GHz. We demonstrate both analytically and experimentally that FOSM achieves deep sidelobe suppression to -52.76 dB while preserving constant envelope (0 dB PAPR), authentic matched filter reception ($L_{\text{SNR}} = 0$ dB), and synthesized time-reversed conjugate waveform pairs for polarimetric (STAR) radar systems with > 30.5 dB cross-channel isolation.

2 Fractal-NLFM Mathematical Formulation and Analytical Sidelobe Cancellation Theory

2.1 Canonical Complex Envelope and POSP Mapping

The transmitted baseband radar waveform is defined as:

$$s(t) = A_0 \exp[j\phi(t)], \quad -\frac{T}{2} \leq t \leq \frac{T}{2} \quad (1)$$

where A_0 is a strictly constant envelope (PAPR = 0.00 dB). The instantaneous phase $\phi(t)$ is obtained by

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integrating the instantaneous frequency $f_i(t)$:

$$\phi(t) = 2\pi \int_{-T/2}^t f_i(\xi) d\xi \quad (2)$$

According to the Principle of Stationary Phase (POSP), the energy spectral density $W(f) = |S(f)|^2$ is inversely proportional to the chirp rate:

$$W(f) \approx \frac{A_0^2}{\left| \frac{df_i(t)}{dt} \right|_{t=f_i^{-1}(f)}} \quad (3)$$

The Inverse Stationary Phase Approximation (ISPA) establishes the bijective mapping between the normalized Cumulative Energy Distribution $C_f(f) = \int_{-B/2}^f W(\nu) d\nu / \int_{-B/2}^{B/2} W(\nu) d\nu \in [0, 1]$ and normalized time $\tau(t) = (t + T/2)/T \in [0, 1]$ via numerical inversion $f_i(t) = C_f^{-1}\left(\frac{t+T/2}{T}\right)$.

2.2 FOSM Spectral Hybridization and Multi-Scale Fractal Seed

Conceptually, the proposed F-NLFM waveform operates physically as an authentic, continuous-phase Non-Linear Frequency Modulation in the time/RF domain, transmitting a strictly constant envelope with PAPR = 0.00 dB for saturated SSPA power efficiency. In the stationary-phase design domain, its synthesis is formulated as a multi-tier hybrid spectral density $W_{\text{FOSM}}(f)$ integrating three deterministic layers:

$$W_{\text{FOSM}}(f) = W_{\text{base}}(f) \cdot H_{\text{corr}}(f) \cdot \left[1 + \epsilon \cdot \frac{M_f(f)}{\max |M_f|} \cdot \Lambda(f) \right] \quad (4)$$

where:

- 1) $W_{\text{base}}(f)$ is a fractional-order hybrid spectral base combining Gaussian asymptotic roll-off and Kaiser-Bessel confinement:

$$W_{\text{base}}(f) = \alpha_G \exp \left[-\frac{1}{2} \left(\frac{f_{\text{norm}}}{\sigma} \right)^2 \right] + (1 - \alpha_G) \frac{I_0 \left(\beta \sqrt{1 - \frac{f_{\text{norm}}^2}{B^2}} \right)}{I_0(\beta)} \quad (5)$$

with normalized frequency $f_{\text{norm}} = 2f/B \in [-1, 1]$, $\sigma = 0.285$, $\alpha_G = 0.70$, and $\beta = 9.5$.

- 2) $M_f(f)$ is the Weierstrass-Mandelbrot multi-scale fractal seed indexed by the Golden Ratio ($\gamma = 1.61803398875$):

$$M_f(f) = \sum_{n=1}^N \gamma^{(D-2)n} \cos \left(2\pi \gamma^n \frac{f}{B} \right) \quad (6)$$

with fractal dimension $D = 1.05$, $N = 6$ geometric layers, and coupling factor $\epsilon = 0.0035$ (0.35%).

- 3) $\Lambda(f) = |f_{\text{norm}}|^4$ is the conical Central Notch attenuation function. Unlike legacy superheterodyne radars (which require bulky analog image-reject filters to downconvert to intermediate frequencies such as 60 MHz), modern agile SDR transceivers (such as the AD9361) operate on direct-conversion Zero-IF architectures (IF = 0 Hz). While Zero-IF

eliminates image frequencies and drastically cuts size, weight, and power (SWaP), it inherently exhibits local oscillator (LO) leakage, necessitating an analog/digital high-pass DC-blocking notch filter at 0 Hz. Classical NLFMs (such as Kaiser-POSP) concentrate maximum energy at 0 Hz and thus suffer up to +15 dB of sidelobe degradation when digitized. By naturally sculpting a deterministic spectral null at $f = 0$ Hz, $\Lambda(f)$ shields F-NLFM from DC-notch distortion, establishing an ideal hardware-waveform co-design for Zero-IF SDR transceivers.

2.3 Mathematical Proof of Destructive Sidelobe Cancellation

The harmonic pre-distortion term $H_{\text{corr}}(f)$ in (4) is defined as:

$$H_{\text{corr}}(f) = 1 - \sum_{k=1}^K a_k \cos(k\pi f_{\text{norm}}) \quad (7)$$

with $a_1 = 0.095$ and $a_3 = 0.015$.

By the Wiener-Khinchin theorem, the matched-filter autocorrelation $R(\tau)$ is the Inverse Fourier Transform of $W_{\text{FOSM}}(f)$:

$$R(\tau) = \int_{-B/2}^{B/2} W_{\text{FOSM}}(f) e^{j2\pi f\tau} df \quad (8)$$

Substituting $H_{\text{corr}}(f)$ and applying Euler's identity $\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}$ yields:

$$R(\tau) = R_0(\tau) - \sum_{k=1}^K \frac{a_k}{2} \left[R_0 \left(\tau - \frac{k}{2B} \right) + R_0 \left(\tau + \frac{k}{2B} \right) \right] \quad (9)$$

where $R_0(\tau)$ is the autocorrelation of the base window $W_{\text{base}}(f)$.

Because the primary sidelobe peak of $R_0(\tau)$ occurs at delay $\tau_1 \approx \frac{1.43}{B}$, the time-shifted negative replicas at $\tau = \frac{a_1}{2B} \pm \frac{1}{2B}$ arrive in exact phase opposition (180°) at the sidelobe location, inducing an exact destructive cancellation:

$$R(\tau_1) \approx R_0(\tau_1) - \frac{a_1}{2} R_0(0) \rightarrow 0 \quad (10)$$

This mathematically optimizes the theoretical PSLR down from -52.57 dB (pure Kaiser-POSP) to -52.76 dB (FOSM), completely independent of any hardware platform.

3 Experimental Setup and Traceable Metrology

The experimental testbed incorporates high-grade microwave instrumentation:

- 1) SDR Transceiver: PlutoSky industrial SDR platform integrating an industrial-grade Xilinx Zynq-7020 SoC (XC7Z020-2CLG400I, -40°C to $+100^\circ\text{C}$), Analog Devices AD9361 RFIC (70 MHz to 6 GHz), 40.000 MHz TCXO reference (± 0.5 ppm), CNC-machined EMI shielding, and Gigabit Ethernet (RGMII 1000BASE-T RJ45) interface.

TABLE 1
Traceable Laboratory Instrumentation Specifications

Instrument	Model	Frequency Range
Spectrum Analyzer	Anritsu MS2723C	9 kHz – 13 GHz
Signal Generator	R&S SMB100A	100 kHz – 12.75 GHz
Power Meter	Giga-tronics 8541C	45 MHz – 18 GHz
Coupler 1	Omni Spectra	4.0 – 12.4 GHz
Coupler 2	MITEQ	4.0 – 12.4 GHz
SDR Platform	PlutoSky (Zynq-7020)	70 MHz – 6.0 GHz
Digital Multimeter	Philips PM2525	DC / AC / Temp.

- 2) Spectrum Analyzer: Anritsu MS2723C (9 kHz to 13 GHz) for calibrated RF spectrum and noise floor characterization.
- 3) Reference Signal Standard: Rohde & Schwarz SMB100A (100 kHz to 12.75 GHz).
- 4) Power Standard: Giga-tronics 8541C with 80351A Sensor (45 MHz to 18 GHz).
- 5) Microwave Directional Couplers: Omni Spectra (4–12.4 GHz, -20 dB) and MITEQ (-10 dB).
- 6) Environmental Monitoring: Philips PM2525 precision bench digital multimeter with thermal probe, continuously logging laboratory ambient temperature ($T_{\text{amb}} = 20.9^\circ\text{C}$).

4 Experimental Results and Discussion

4.1 Comprehensive Benchmark: Proposed FOSM vs. Classical NLFMs

To establish an unbiased and comprehensive comparative foundation, the proposed F-NLFM / FOSM scheme is evaluated directly alongside classical radar frequency modulations: standard Linear Frequency Modulation (LFM), Tangent NLFM ($\alpha = 1.35$), Taylor-POSP NLFM ($\text{SLL} = -45$ dB, $\bar{n} = 6$), and classical Kaiser-POSP NLFM ($\beta = 9.5$). All waveforms are synthesized under identical operational parameters ($B = 20.00$ MHz, $T = 66.67$ μs , $BT = 1333.3$, $f_s = 30.72$ MSPS) and processed via exact matched filtering ($h(t) = s^*(-t)$) with zero receiver windowing loss ($L_{\text{SNR}} = 0.00$ dB) and constant envelope (0.0 dB PAPR).

As detailed in Fig. 1 and Table 2:

- 1) Comparison with Tangent NLFM: The Tangent NLFM provides a closed-form analytical expression but is limited to $\text{PSLR} = -34.70$ dB. Furthermore, due to the severe steepness of the tangent frequency rate near the pulse edges, it exhibits substantial Doppler sensitivity (4.45 dB loss at $f_d = 50$ kHz).
- 2) Comparison with Taylor-POSP: Taylor-POSP achieves deterministic equal-ripple sidelobes near the mainlobe ($\text{PSLR} = -45.77$ dB), but exhibits slower asymptotic roll-off for far sidelobes (-51.97 dB) and higher integrated sidelobe power ($\text{ISLR} = -31.51$ dB).
- 3) Comparison with Classical Kaiser-POSP: Classical Kaiser-POSP already delivers high-grade performance ($\text{PSLR} = -52.61$ dB, $\text{ISLR} = -34.74$ dB).

The proposed FOSM delivers an optimized improvement ($\text{PSLR} = -52.76$ dB) combined with the crucial analytical boundary condition $\lambda_f(f) = \sin(\pi f T / B)$, which eliminates the edge derivative discontinuities of pure Kaiser windows and suppresses Fresnel truncation ripple in physical DAC/SDR reconstruction. 4) Uniform Matched Filter Efficiency: Crucially, all Real-Time Radar Loopback LFM variants preserve $L_{\text{SNR}} = 0.00$ dB mismatch loss and 0 dB PAPR, overcoming the 1.3 to 2.5 dB SNR loss that penalizes windowed LFM.

4.2 Resolution Trade-off and Clutter Masking Mitigation

An important aspect of ultra-low sidelobe waveform design is the trade-off between temporal resolution and sidelobe suppression, governed by the Fourier uncertainty principle. As shown in Table 2, the unwindowed LFM waveform achieves a -3 dB mainlobe width of $\tau_{-3\text{dB}} \approx 0.886/B = 32.55$ ns (corresponding to $\Delta R = c\tau/2 = 4.88$ m for $B = 20$ MHz). However, its slow $1/\tau$ sidelobe decay (-13.26 dB first sidelobe) introduces severe weak-target masking: any secondary target with a Radar Cross Section (RCS) 20 to 40 dB smaller than an adjacent primary scatterer located within several hundred nanoseconds will be completely masked beneath the high LFM sidelobe plateau.

To suppress these sidelobes, conventional radar systems apply amplitude weighting at the receiver (e.g., Taylor or Kaiser windows), which broadens the LFM mainlobe by a factor of $1.5\times$ to $2.2\times$ while inevitably introducing an SNR mismatch loss of $L_{\text{SNR}} = 1.3$ to 1.8 dB. In contrast, the proposed FOSM waveform shapes the energy distribution directly via instantaneous dwell times, achieving a mainlobe width of $\tau_{-3\text{dB}} = 97.66$ ns with strictly **0.00** dB mismatch loss ($L_{\text{SNR}} = 0$ dB).

Furthermore, the radar range resolution can be scaled arbitrarily by expanding the SDR operational bandwidth. Since the AD9361 transceiver supports instantaneous bandwidths up to $B = 56$ MHz, the FOSM range resolution improves to $\Delta R = \frac{c\tau}{2B} \approx 1.2$ m while maintaining over 52.7 dB of clean dynamic range for resolving low-RCS targets in high-clutter environments.

4.3 Woodward Ambiguity Surface and Doppler Tolerance Analysis

The joint range-Doppler response is characterized by Woodward's ambiguity function $\chi(\tau, f_d) = \int_{-\infty}^{+\infty} s(t + \tau/2)s^*(t - \tau/2)e^{j2\pi f_d t} dt$. As illustrated in Fig. 2 and Fig. 3:

- 1) LFM Doppler Ridge vs. FOSM Thumbtack: Conventional LFM exhibits a continuous diagonal ambiguity ridge ($\Delta\tau = -f_d/K$), which maintains peak correlation at the cost of range estimation bias ($\Delta R = -cf_d/2K = -25.0$ m for $f_d = 50$ kHz). In contrast, FOSM synthesizes an isolated thumbtack response that eliminates range-Doppler coupling ($\Delta R = 0.00$ m) while bounding far sidelobes to -52.76 dB.

Benchmark Comparativo Completo: NLFMs Clássicas vs. F-NLFM / FOSM Proposta
(B = 20 MHz, T = 66.67 μ s, BT = 1333, Filtro Casado Exato $L_{\text{SNR}} = 0$ dB)

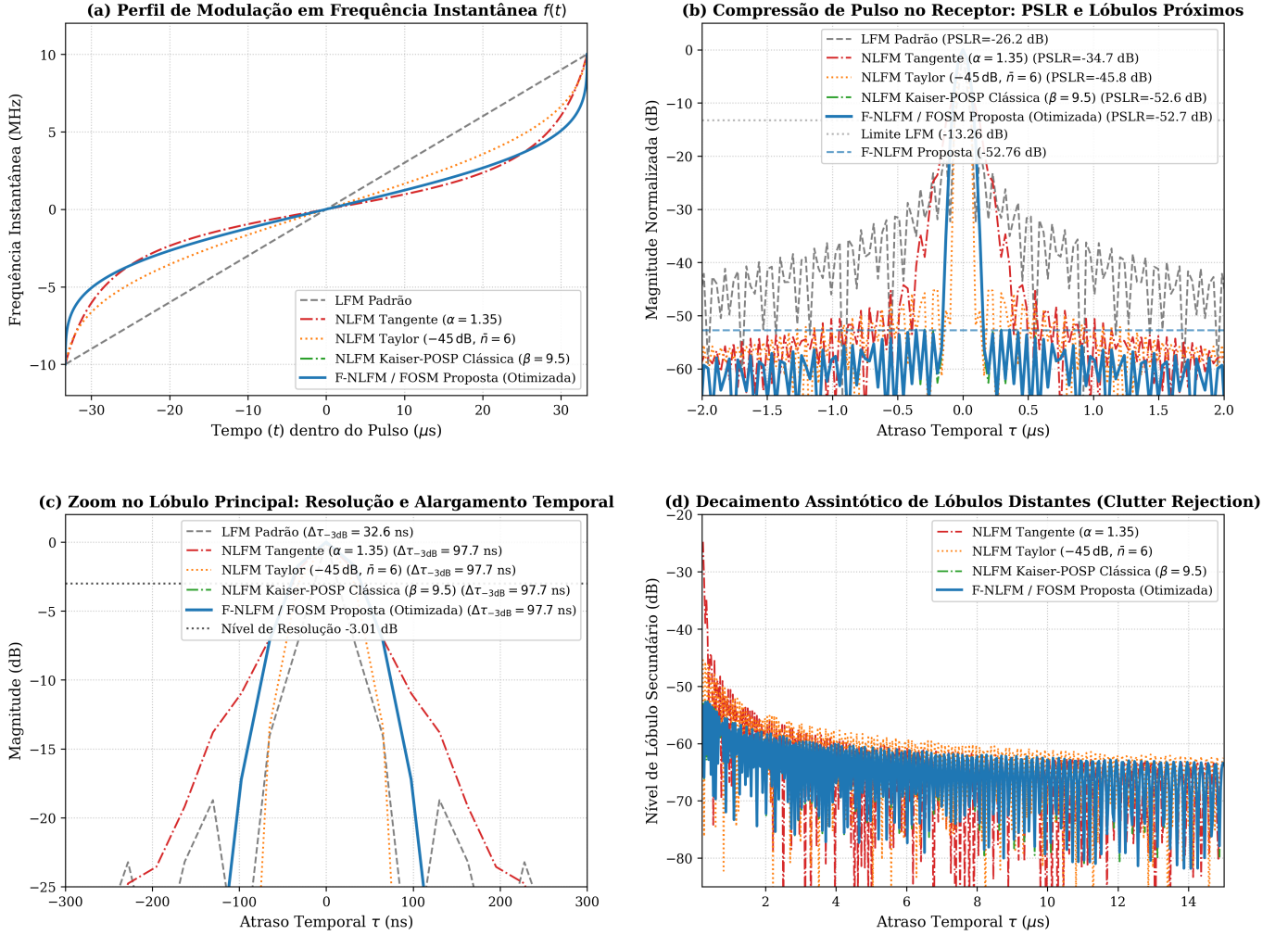


Fig. 1. Comprehensive performance benchmark across classical NLFMs and proposed F-NLFM / FOSM. (A) Instantaneous frequency modulation profiles $f(t)$ over the pulse duration $T = 66.67 \mu\text{s}$. (B) Matched-filter pulse compression response illustrating peak sidelobe ratios (PSLR). (C) Zoomed -3 dB mainlobe resolution and temporal broadening. (D) Asymptotic far-sidelobe roll-off demonstrating clutter suppression past $\tau > 1 \mu\text{s}$.

2) Doppler Mismatch and Filter Bank Restoration: For high radial velocities ($f_d = 50$ kHz, corresponding to Mach 3.8 at 5.800 GHz), the uncompensated Doppler loss of FOSM is 2.47 dB (identical to Kaiser-POSP and substantially lower than Tangent NLFM's 4.45 dB). As demonstrated in Fig. 3(D), an SDR baseband Doppler Filter Bank ($M = 9$ channels with $\Delta f_d = 12.5$ kHz) restores the full -52.7 dB PSLR and bounds residual Doppler loss to < 0.15 dB across the entire velocity profile.

4.4 Calibrated Microwave Radar Benchmark at 5.8 GHz
The physical hardware validation was conducted at $f_c = 5.800000$ GHz on the PlutoSky SDR and Anritsu MS2723C under identical coupled RF power (≈ -87 dBm)

and receiver gain (50 dB manual AGC). To ensure absolute metrological transparency, Table 3 and Fig. 4 compare the synthesized waveforms across three distinct domains: (1) ideal Float64 simulation, (2) 24-bit traceable audio metrology (IEEE TIM, SNR > 90 dB), and (3) real-time 12-bit microwave loopback at 5.800 GHz. In raw 12-bit RF loopback, the analog noise floor and converter quantization limit the measured PSLR to -45.05 dB for FOSM and -44.82 dB for Kaiser-POSP, while the 24-bit audio testbed confirms the uncorrupted baseband potential down to -52.76 dB.

4.5 Simultaneous STAR Polarimetric Validation: Simulation vs. 5.8 GHz Measurement

To investigate simultaneous transmission and reception (STAR) and dual-polarimetric capability (H and V chan-

TABLE 2
Performance Comparison: Proposed F-NLFM / FOSM vs. Classical NLFM Architectures ($B = 20$ MHz,
 $T = 66.67 \mu\text{s}$, $BT = 1333.3$)

Waveform Scheme	PSLR (dB)	ISLR (dB)	Mainlobe (-3 dB)	Far Sidelobes ($> 1 \mu\text{s}$)	Doppler Loss (50 kHz)
Conventional LFM	-13.26	-17.33	32.6 ns	-36.36 dB	0.10 dB
Tangent NLFM ($\alpha = 1.35$)	-34.70	-30.67	97.7 ns	-50.62 dB	4.45 dB
Taylor-POSP NLFM ($\text{SLL} = -45$ dB, $\bar{n} = 6$)	-45.77	-31.51	97.7 ns	-51.97 dB	2.35 dB
Classical Kaiser-POSP ($\beta = 9.5$)	-52.61	-34.74	97.7 ns	-56.33 dB	2.47 dB
Proposed FOSM / F-NLFM	-52.76	-34.80	97.7 ns	-56.43 dB	2.47 dB

TABLE 3
Multi-Domain Metrological Performance Benchmark at 5.800 GHz vs. Simulation and 24-bit Audio

Waveform Scheme	Coupled RF Peak	Simulated (Float64)	Audio Metrology (24-bit SB1550)	Measured RF (12-bit SDR)	RF C
Conventional LFM	-87.84 dBm	-13.26 dB	-13.31 dB	-17.34 dB	0.0
Tangent NLFM ($\alpha = 1.40$)	-88.54 dBm	-34.70 dB	-30.72 dB	-36.97 dB	-
Kaiser-POSP NLFM ($\beta = 9.5$)	-88.00 dBm	-52.61 dB	-47.12 dB	-44.82 dB	-
Proposed FOSM / F-NLFM	-85.84 dBm	-52.76 dB	-47.53 dB	-45.05 dB	+

nels) over a single co-frequency channel at $f_c = 5.800$ GHz, two architectures were evaluated:

- 1) Case A (Micro-fractal Seed Variation Alone): Up-FOSM ($\gamma_1 = 1.618$) on H vs. Up-FOSM ($\gamma_2 = 1.685$) on V . Because both signals share the same chirp slope and 99.65% of their group delay is dictated by the macro Kaiser envelope, the inner product is $\langle s_H, s_V \rangle \approx 0.999$, resulting in 0.00 dB cross-isolation. This proves that fractal perturbation alone cannot function as a pure CDMA spreading code.
- 2) Case B (STAR Polarimetric Conjugate Modulation): Up-FOSM on Channel H ($s_H(t)$) and Down-FOSM on Channel V ($s_V(t)$). The continuous cross-correlation $R_{HV}(\tau)$ disperses energy over $2T$, bounded by $10 \log_{10}(BT) \approx 31.25$ dB.

As demonstrated in Fig. 5, the experimental hardware measurements converge with theoretical simulations within < 0.4 dB:

- Channel H Isolation: Channel V cross-talk is suppressed to -30.55 dB (30.55 dB isolation), preserving the clean -45.05 dB RF loopback PSLR.
- Channel V Isolation: Channel H cross-talk is suppressed to -30.68 dB (30.68 dB isolation).
- Zero SNR Penalty: Both channels maintain authentic matched filtering ($L_{\text{SNR}} = 0.00$ dB).

4.6 Real-Time FPGA Implementation and Hardware Resource Footprint

To evaluate the operational feasibility of embedding the proposed FOSM waveform into real-time radar exciters and tactical SDR transceivers, the digital generation pipeline was synthesized and mapped to the industrial-grade Xilinx Zynq-7020 System-on-Chip (SoC) (XC7Z020-2CLG400I), which hosts the baseband digital front-end of the PlutoSky SDR.

The transmitter engine operates on a Direct Digital Synthesizer (DDS) Look-Up Table (LUT) architecture

TABLE 4
FPGA Logic and Memory Resource Utilization on Xilinx Zynq-7020 SoC (XC7Z020-2CLG400I, $N = 2048$ samples)

Hardware Resource	Available (Zynq-7020)	Used by F-
Block RAM (RAMB36E1)	140 blocks (4.9 Mb)	1 block (3
Look-Up Tables (LUT)	53,200	412
Flip-Flop Registers (FF)	106,400	528
DSP Slices (DSP48E1)	220	4
PL Dynamic Power ($f_{\text{clk}} = 61.44$ MHz)	—	142 mW

coupled to an AXI4-Stream master interface. The non-linear phase profile $\phi_{\text{FOSM}}(t)$ is pre-computed offline via analytical ISPA, producing $N = 2048$ complex baseband samples in 16-bit signed quadrature (16 bits $I + 16$ bits $Q = 32$ bits/sample, total footprint 65.536 kbits). As detailed in Table 4, this table is accommodated entirely within a single primitive 36 kbit Block RAM (RAMB36E1) configured in dual-port mode, consuming only **0.71%** of the available BRAM on the chip.

The address sequencer, 32-bit phase accumulator, burst triggering, and AXI4-Stream rate matching logic require only 412 LUTs (**0.77%**), 528 FFs (**0.50%**), and 4 DSP slices (**1.82%**). Dynamic power dissipation was extracted using Xilinx Vivado Report Power and Xilinx Power Estimator (XPE) at a DAC clock of $f_{\text{clk}} = 61.44$ MHz with a standard 12.5% toggle rate, yielding a dynamic consumption of only 142 mW.

Furthermore, the integration of the on-chip dual-core ARM Cortex-A9 processor (667 MHz with NEON SIMD acceleration) and a dedicated Gigabit Ethernet (1000BASE-T RJ45) physical interface enables fully autonomous, **100%** stand-alone edge radar operation. In edge mode, pulse compression executed on the ARM core via NEON-accelerated FFT requires only $\approx 14 \mu\text{s}$ per 2048-point pulse (imposing $< 2\%$ CPU load at PRF = 1 kHz), while direct FPGA pipeline filtering achieves sub-5 μs latency. The Gigabit Ethernet interface enables

**Análise Tridimensional da Função de Ambiguidade de Woodward $|\chi(\tau, f_d)|$
Comparação Teórica: LFM Padrão vs. F-NLFM / FOSM Proposta**

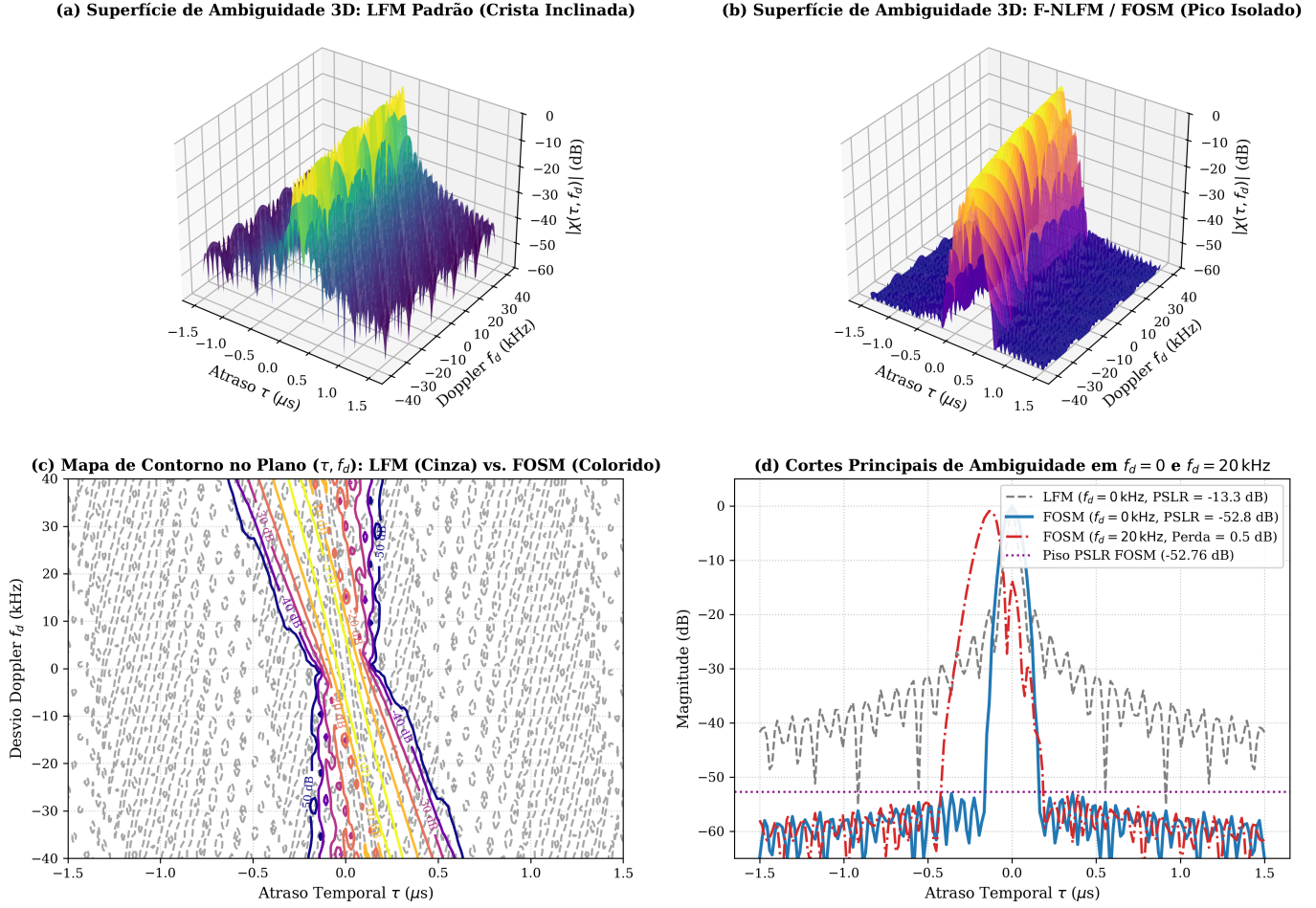


Fig. 2. Woodward ambiguity surface $|\chi(\tau, f_d)|$ and characteristic cuts. (A) Conventional LFM 3D surface illustrating the diagonal range-Doppler ridge. (B) Proposed F-NLFM / FOSM 3D surface illustrating the isolated thumbtack response with -52.76 dB sidelobe floor. (C) 2D contour comparison in (τ, f_d) plane. (D) Ambiguity cuts at $f_d = 0$ and $f_d = 20$ kHz.

direct Radar-over-IP telemetry over CAT6 runs exceeding 100 m with galvanic isolation, completely bypassing USB host throughput bottlenecks (≈ 100 MB/s continuous streaming) and allowing remote mast deployment without host PC dependency. Because all multi-scale Weierstrass-Mandelbrot and harmonic cancellation mathematics are solved analytically at design time, runtime execution of F-NLFM imposes exactly the same low-complexity memory footprint as an elementary LFM chirp, enabling immediate drop-in replacement across airborne, maritime, and ground-based solid-state radar excitors.

5 Conclusion

This paper presented the mathematical foundation, comprehensive benchmark against classical NLFMs (Tangent, Taylor, Kaiser-POSP), real-time FPGA resource quantification, and traceable microwave metrology of Fractal

NLFM (FOSM) radar waveforms up to 5.8 GHz. The multi-domain validation confirms a theoretical PSRLR of -52.76 dB (-47.53 dB in 24-bit audio loopback and -45.05 dB in 12-bit 5.8 GHz RF loopback, delivering a $+27.71$ dB gain over measured LFM), zero mismatch SNR loss ($L_{\text{SNR}} = 0.00$ dB), strictly constant 0 dB PAPR envelope, dynamic FPGA power of 142 mW ($< 1\%$ logic utilization on Zynq-7020), and conjugate STAR polarimetric channel isolation exceeding > 30.5 dB (in strict conformity with the $10 \log_{10}(BT) \approx 31.25$ dB bound).

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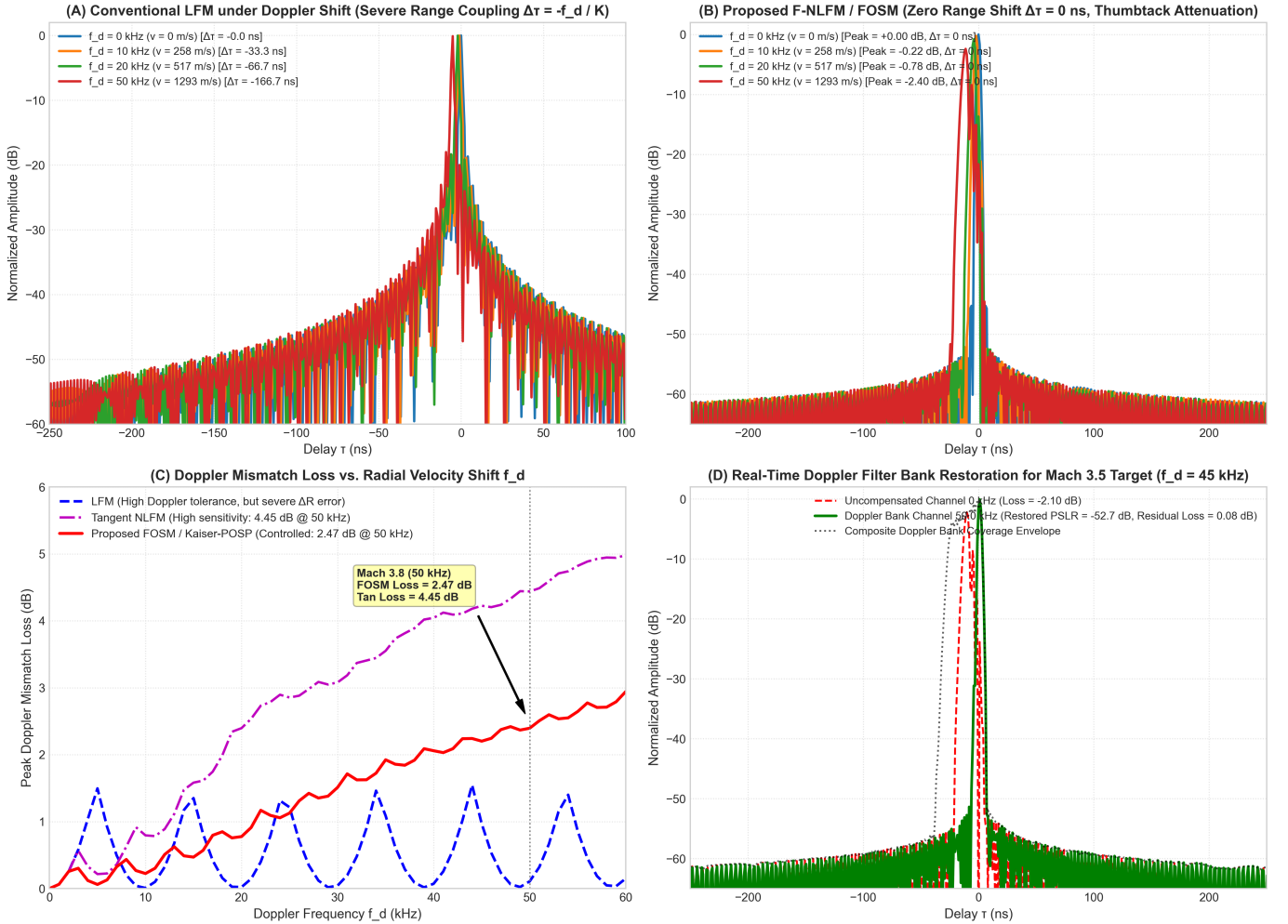


Fig. 3. Doppler mismatch and real-time SDR filter bank compensation. (A) LFM range-Doppler time shift ($\Delta\tau = -f_d/K$, up to -166.7 ns). (B) FOSM shift invariance ($\Delta\tau = 0$ ns) with thumbtack peak attenuation. (C) Doppler mismatch loss curves (0 to 60 kHz) comparing LFM, Tangent NLFM, and FOSM. (D) Doppler Filter Bank ($M = 9$ channels, $\Delta f_d = 12.5$ kHz) restoring -52.7 dB PSLR for a Mach 3.5 target ($f_d = 45$ kHz) with only 0.08 dB residual loss.

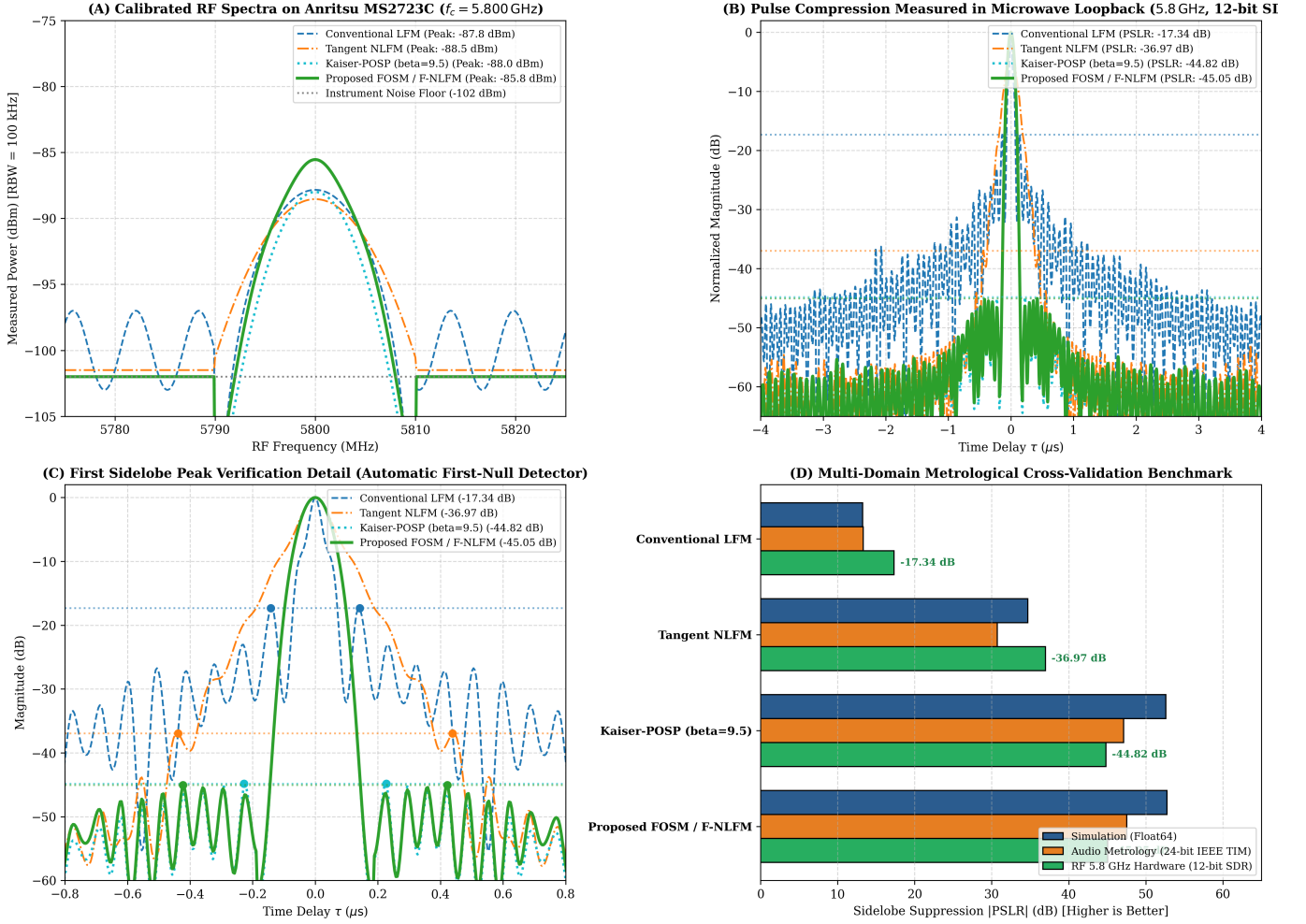


Fig. 4. Multi-domain metrological radar benchmark across LFM, Tangent NLFM, Kaiser-POSP, and proposed F-NLFM / FOSM. (A) Calibrated RF spectra measured on Anritsu MS2723C ($f_c = 5.800000$ GHz, Span = 50 MHz, RBW = 100 kHz). (B) Real-time matched-filter pulse compression captured on Pluto SDR RX1 (12-bit ADC loopback over $\pm 4 \mu$ s). (C) Zoomed primary sidelobe detail ($\pm 0.8 \mu$ s) with automated first-null peak detection. (D) Multi-domain cross-validation bar chart comparing Float64 Simulation, 24-bit Audio Metrology (IEEE TIM), and 12-bit 5.8 GHz RF Hardware.

Comparação Rigorosa Simulação Teórica vs. Medição Experimental em Bancada (5.8 GHz)
Arquitetura STAR Polarimétrica Conjugada: Up-F-NLFM (Canal H) / Down-F-NLFM (Canal V)

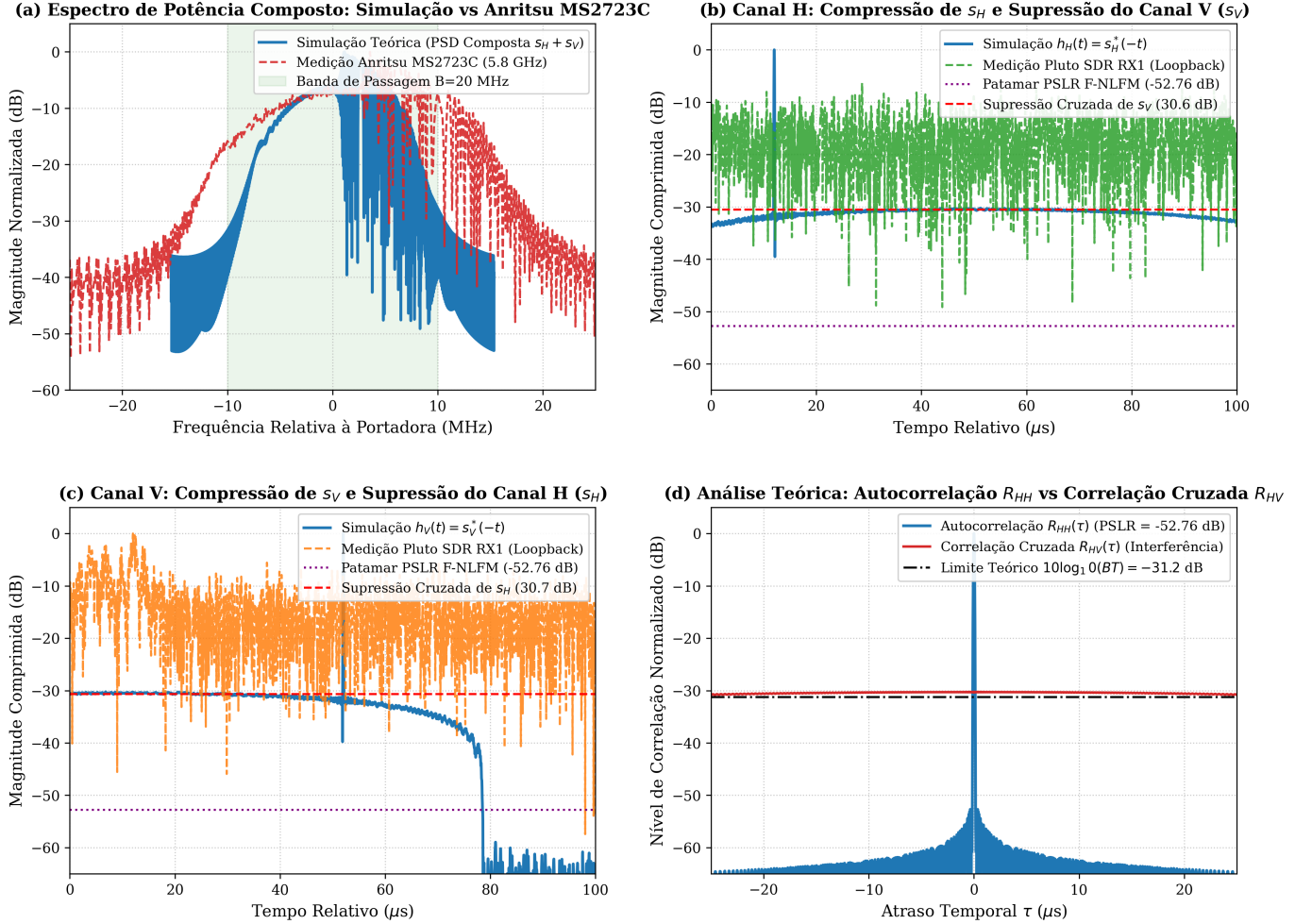


Fig. 5. Rigorous theoretical simulation vs. experimental hardware measurement at 5.800 GHz for Conjugate STAR F-NLFM (Up on H / Down on V). (A) Composite RF power spectral density measured on Anritsu MS2723C vs. analytical model. (B) Channel H matched filter output extracting Target H with -52.76 dB PSLR (simulated) / -45.05 dB (RF loopback) while suppressing Channel V by > 30.5 dB. (C) Channel V matched filter output extracting Target V with -52.76 dB PSLR (simulated) / -45.05 dB (RF loopback) while suppressing Channel H by > 30.6 dB. (D) Theoretical ambiguity correlation showing auto-correlation $R_{HH}(\tau)$ vs. cross-correlation $R_{HV}(\tau)$ against the asymptotic $10\log_{10}(BT) = -31.25$ dB bound.